

Rocksdb_dataset-5

Commit 81:

a10b4aa9a38571d992b5588a2f08369bc1e19b18

Non-C file modifications, therefore no performance regression testing is required.

Commit 82:

7f271a3fa86825d36029305e47d1217e5e5364e2

The 1st symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2012646.4865102042
4  The new version is executed by default
5  1967415.1616874998
6  The new version
7  2043749.560734694
8  Comparison of New and old versions (along symbolic execution path)
9  -0.015218632860920936
10 Comparison of New and old versions (along the default execution path)
11 -0.03735017269912127
```

No performance regression

The 2nd symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2032903.1457916666
4  The new version is executed by default
5  1978396.4269375
6  The new version
7  1983479.996638298
8  Comparison of New and old versions (along symbolic execution path)
9  0.0249173922787895
10 Comparison of New and old versions (along the default execution path)
11 -0.002562954861865968
```

No performance regression

The 3rd symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2007528.0237291667
4  The new version is executed by default
5  1993451.6437291668
6  The new version
7  1998855.25800000001
8  Comparison of New and old versions (along symbolic execution path)
9  0.004338866305829617
10 Comparison of New and old versions (along the default execution path)
11 -0.0027033544571106313
```

No performance regression

Commit 83:

7774a4de17cebf74de7898a46d3ece7287e2c6f1

Non-C file modifications, therefore no performance regression testing is required.

Commit 84:

864964b008e96dfe69041f487c8a47cea1ab61b0

Unable to provide test cases

Commit 85:

22270dea6fb7fb050de0ad44e9d42fd355e408fb

Non-C file modifications, therefore no performance regression testing is required.

Commit 86: f5f4f83fcc9af320203a8974c47e0aaac6b3ebe0

Non-C file modifications, therefore no performance regression testing is required.

Commit 87:

61ee80faa01828d33bf218d06b981da60d401c0b

The 1st symbolic execution

Testcase:for(int i = 0; i< 1000; i++){db->SetOptions({{"write_buffer_size", "131072"}});}

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  838899.8141666666
4  The new version is executed by default
5  833938.3052340427
6  The new version
7  851723.4713333333
8  Comparison of New and old versions (along symbolic execution path)
9  -0.01505612748535844
10 Comparison of New and old versions (along the default execution path)
11 -0.02088138544714372
```

No performance regression

Commit 88:

242e69f0670118cd0def618a15f3c571e9001a77

Non-C file modifications, therefore no performance regression testing is required.

Commit 89:

ed2a87db07b67ba402914d69edad5d528f2a0370

Non-C file modifications, therefore no performance regression testing is required.

Commit 90:

833a2266a394fe5f140d2a22f406c82bb605c726

Unable to provide test cases

Commit 91:

4ace09f6eb60ced939cb15e2340d0a038c56a465

Non-C file modifications, therefore no performance regression testing is required.

Commit 92:

354025fc862d7782c4509feb0b29fc5823a5447e

Unable to provide test cases

Commit 93:

62531da51027bbebf3ae7d51399968d4043da49b

Unable to provide test cases

Commit 94:

302254d928402c80ecabb0cf2b76d18be35a87b9

Non-C file modifications, therefore no performance regression testing is required.

Commit 95: c1fb33e1d07a30f6f1ef14a3e7c65020838c33cc

The 1st symbolic execution

```
Testcase:for (int i = 0; i < 100; i++){rocksdb::BackupEngine
*backup_engine;rocksdb::BackupEngine::Open(rocksdb::Env::Default(),
rocksdb::BackupEngineOptions("/tmp/test"), &backup_engine);backup_engine-
>CreateNewBackup(db);delete backup_engine;}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  275084.744580000006
4  The new version is executed by default
5  267115.7775102041
6  The new version
7  264884.90232
8  Comparison of New and old versions (along symbolic execution path)
9  0.038506695438903955
10 Comparison of New and old versions (along the default execution path)
11 0.00842205490258206
```

No performance regression

Commit 96:

a377bded9f03fd34db8ca36bc4e00b323c7ed5df

Non-C file modifications, therefore no performance regression testing is required.

Commit 97:

dd01f73e26978de0733d2bbac041230a3631df15

Unable to provide test cases

Commit 98:

d48af213860054a7696e7ea2764f266c88a3263e

The 1st symbolic execution

```
Testcase:for(int i = 0; i < 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db-
>Put(rocksdb::WriteOptions(), key, value2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2031581.0888297874
4  The new version is executed by default
5  2021861.7380625
6  The new version
7  2032756.462625
8  Comparison of New and old versions (along symbolic execution path)
9  -0.0005782167302494567
10 Comparison of New and old versions (along the default execution path)
11 -0.005359581810617393
```

No performance regression

Commit 99:

1f36399a77405bb9b5ff9416db039c6493a38ec4

Non-C file modifications, therefore no performance regression testing is required.

Commit 100: c9ce4a3d6b796fff1e5d2f7f865f3467d056fccc

The 1st symbolic execution

Testcase:for (int i = 0; i < 10000; i++){key = "key_exit" + std::to_string(i + 30001);value = "new_value" + key;db->Put(write_options_sync, key, value);}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  348679.52379166667
4  The new version is executed by default
5  320448.890367347
6  The new version
7  316206.50228571426
8  Comparison of New and old versions (along symbolic execution path)
```

```
9 0.10269561590675579
10 Comparison of New and old versions (along the default execution path)
11 0.013416511206968922
```

No performance regression

The 2nd symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 2029535.2152978724
4 The new version is executed by default
5 2036983.8432500002
6 The new version
7 2027447.8026249998
8 Comparison of New and old versions (along symbolic execution path)
9 0.0010295765297483944
10 Comparison of New and old versions (along the default execution path)
11 0.004703470349596061
```

No performance regression

The 3rd symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 2030193.6210625002
4 The new version is executed by default
5 2020669.1581666667
```

```
6 The new version
7 2030272.4297142855
8 Comparison of New and old versions (along symbolic execution path)
9 -3.881678666958835e-05
10 Comparison of New and old versions (along the default execution path)
11 -0.004730040859083246
```

No performance regression

The 4th symbolic execution

Testcase:for(int i = 0; i< 1000; i++){db->SetOptions(("{\"write_buffer_size", "131072"}));}

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 679645.5725000001
4 The new version is executed by default
5 672378.5174042553
6 The new version
7 821731.0792916667
8 Comparison of New and old versions (along symbolic execution path)
9 -0.17290998280622963
10 Comparison of New and old versions (along the default execution path)
11 -0.18175357565415867
```

No performance regression